

SOLVED QUESTIONS SETS

SET-A

1. **TCP friendly rate control (TFRC) is an alternative to:**
 - A. TCP/IP
 - B. UDP
 - C. Multiplexing and de- multiplexing
 - D. Congestion control
2. **A data packet is a packet header together with:**
 - A. A network layer
 - B. User data
 - C. An administrative layer
 - D. A packet switches
3. **Which of the following is involved in NFS?**
 - A. RPC
 - B. FTP
 - C. TCP/IP
 - D. RPC & TCP/IP
4. **ARP is an abbreviation of:**
 - A. Application resolution protocol
 - B. Access resolution protocol
 - C. Address resolution protocol
 - D. Active resolution protocol
5. **The amount of uncertainty in a system of symbol is called?**
 - A. Band with
 - B. Loss
 - C. Entropy
 - D. Quantum
6. **Sending messages to well defined groups that are numerically large in size but small compared to the network as a whole is called**
 - A. Broadcasting
 - B. Multicasting
 - C. Flooding
 - D. Routing
7. **"To access a complex object in a recursive way building the object from other objects" is an example of which pattern?**
 - A. Recursive
 - B. Recursive composite
 - C. Composite
 - D. Abstract
8. **A friend function can be used to:**
 - A. Avoid arguments between classes
 - B. Allow access to classes whose source code is unavailable
 - C. Allow one class to access an unrelated class
 - D. Increase the versatility of an over loaded operator
9. **An expression contains relational operators' assignment operators and arithmetic operators. In the absence of parentheses, they will be evaluated in which of the following order:**
 - A. Assignment, relational, arithmetic
 - B. Relational, arithmetic, assignment
 - C. Arithmetic, Relational, Assignment
 - D. Assignment, arithmetic, relational
10. **In object-oriented, polymorphism means:**
 - A. There can be many objects in the design
 - B. Methods can be changed in many ways
 - C. Objects can implement the same method in many ways
 - D. None of the above

11. How many values can be returned from function?
A. 0 B. 1
C. 2 D. 3
12. The scope resolution operator usually:
A. Limits the visibility of variables to a certain function
B. Tells what base class is derived from
C. Specifies a particular class
D. Providing a useful conceptual framework
13. This is a system of program and data structure that approximate the approach of the human brain:
A. Intelligent network
B. Neural network
C. Decision support system
D. Genetic programming
14. Which of the following search algorithm is not heuristic search?
A. Greedy search
B. A*
C. Iterative deepening
D. Beam search
15. In learning the condition action component, the agent receives some evaluation of its action but is not told the correct action, this is called:
A. Supervised learning
B. Reinforcement learning
C. Unsupervised learning
D. Feedback learning
16. Name the sort for which time is not proportional to n^2 : -
A. Selection sort B. Quick sort
C. Bubble sort D. Insertion sort
17. Given two sorted lists of size 'm' and 'n' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be:
A. m, n B. $\min(m, n)$
C. $\max(m, n)$ D. $m + n - 1$
18. A linear list in which elements can be added or removed at either end but not in the middle, is known as:
A. Queue B. Stack
C. De queue D. Tree
19. Which of the following sorting algorithms does not have a worst-case running time of $O(n^2)$?
A. Insertion sort B. Quick sort
C. Merge sort D. Bubble sort
20. A graph consists of:
A. Nodes B. Arrows
C. Arcs D. All of the above
21. Which approach(s) solve problems by combining the solutions to sub problems?
A. Dynamic programming
B. Merge sort
C. Divide and conquer method
D. All of the above
22. Which is not a practical organization in Flynn's classification of multiprocessor systems?
A. Single instruction single data (SISD)
B. Single instruction multiple data (SIMD)
C. Multiple instruction single data (MISD)
D. Multiple instruction multiple data (MIMD)
23. Intel 8088 microprocessor has got:
A. Four-byte instruction queue
B. Eight-byte instruction queue
C. Six-byte instruction queue
D. Sixteen-byte instruction queue

24. A Winchester disk is a:
 A. Disk stack
 B. Flexible disk
 C. Removable disk
 D. All of the above
25. Which architecture allows instruction and data to be mixed and sorted in the same memory module:
 A. Harvard architecture
 B. RISC architecture
 C. The Von Neumann architecture
 D. None of the above
26. In which addressing mode, the effective address of the operand is generated by adding a constant value to the contents of a register?
 A. Absolute mode B. Index mode
 C. Indirect mode D. Immediate mode
27. Micro program is:
 A. Name of source program in micro computers
 B. The set of instructions indicating the primitive operations in a system
 C. Primitive form of macros used in assembly language programming
 D. Program of very small size
28. A combinational circuit of a full-adder that performs arithmetic sum of two input bits x and y and third input Z , which represents the carry from previous lower significant position, has output carry C as:
 A. $xy + yz$ B. xyz
 C. $x + y + z$ D. $xy + xz + yz$
29. The number of columns in a state table for a sequential circuit with ' m ' flip-flops and ' n ' inputs is:
 A. $M+n$ B. $2m+n$
 C. $M+2n$ D. $2m+2n$
30. $(r-1)$'s complement of the $(r-1)$'s complement of an integer number is:
 A. The original number
 B. The r 's complement
 C. The $(r-2)$'s complement of the number
 D. None of the above
31. Which logic family dissipates the minimum power?
 A. DTL
 B. ECL
 C. TTL
 D. CMOS
32. The logic micro operations can:
 A. Change bit values
 B. Clear a group of bits
 C. Insert new bit values into a register
 D. All of the above
33. Alpha testing is an:
 A. Incremental integration testing
 B. Unit testing
 C. Acceptance testing conducted at developer's site by a customer
 D. Acceptance testing conducted at one or more customers sites by the end users
34. Software configuration management is:
 A. Used in analysis
 B. An umbrella activity
 C. Used in testing phase only
 D. None of the above
35. Maintenance has the following division:
 A. Perfective evolvability
 B. Corrective evolvability
 C. Adaptive, reparability
 D. All of the above

36. Which of the following might be output as a result of using a CASE tool?
- Prototypes
 - Programming code
 - Cost/benefit analysis
 - All of the above
37. Which of the following is not a phase in the SDLC?
- Define the latest technology
 - Implement new system
 - Design new system
 - Analysis current system
38. The process of analyzing a program in an effort to create a representation of the program at a higher level of abstraction than source code is called:
- Forward engineering
 - Reverse programming
 - Code restructuring
 - None of the above
39. The tool that interacts with an executing program, checking path coverage, testing assertion about the value of specific variables and otherwise instrumenting the execution flow of the program is called....
- Test management tools
 - Static analysis tools
 - Dynamic analysis tools
 - Programming code
40. One approach to standardizing storing of data is:
- MIS
 - CODASYL specifications
 - Structured programming
 - All of the above
41. A form can be used to:
- Modify records
 - Format printed
 - Delete records
 - All of the above
42. GO BOTTOM and SKIP-3 commands given one after another in a database file of 30 records sift the control to the:
- 28th record
 - 3rd record
 - 27th record
 - 4th record
43. Database system has 3 layers. These are:
- Physical, logical, view level
 - Logical, conceptual, view level
 - Physical, logical, conceptual level
 - Domain specific, logical, view level
44. Which of the following is not a logical data-base structure?
- Chain
 - Network
 - Tree
 - Relational
45. The disadvantage of shadow paging approach is that:
- It is hard to extend it to allow multiple transactions to execute separately
 - It requires less disk access than log-based recovery approach
 - It requires less disk access than log-based recovery approach
 - It is hard to extend it to allow multiple transactions to execute concurrently
46. FIFO scheduling is:
- Preemptive scheduling
 - Deadline scheduling
 - Non-preemptive scheduling
 - Fair-share scheduling

47. What is the initial value of the semaphore to allow only one of the many processes to enter their section?
 A. 0 B. 2
 C. 1 D. 4
48. Moving process from main memory to disk is called...
 A. Swapping B. Caching
 C. Scheduling D. Spooling
49. A compiler is a program that:
 A. Places programs into memory and prepares them for execution
 B. Automate the translation of assembly language into machine language
 C. Accepts a program written in a high-level language and produces an object program
 D. None of the above
50. The main function of the directory system is to map the:
 A. Binary name of the file onto the information needed to locate the data
 B. ASCII name of the file onto the information needed to locate the data
 C. I-node number of the file with the disk address needed to locate the data
 D. A and B of the above
51. An operating system with multiprogramming capability is one that:
 A. Allows several users to use the same program at once by giving intelligence
 B. Run two or more programs at the same time executing them as needed
 C. Loads several independent processes into memory and switches the CPU from one job to another as required
 D. None of the above.
52. Which of the following is not decidable?
 A. Given a Turing machine M , a string s and an integer k , M accepts s within k steps
 B. Equivalence of two given Turing machine
 C. Language accepted by a given finite state machine is not empty
 D. Language generated by a context
53. For which of the following application regular expression can't be used....
 A. Designing compilers
 B. Simulating sequential circuits
 C. Developing text editor
 D. All of the above
54. Which of the following measures the sum of the series $1 + 2 + 3 + \dots + n$:
 A. $= O(n^2) + O(n)$ B. $= 1/2 n^2 + O(n)$
 C. $= O(n^2)$ D. $= O(n^2) - n$
55. A grammar that produce more than one parse tree for some sentence is called:
 A. Ambiguous
 B. Regular
 C. Unambiguous
 D. None of the above
56. Which one is true?
 A. Stack allocation manages the run-time storage as a stack
 B. Stack allocation manages the compile-time storage as a stack
 C. Heap allocation manages both the run-time and compile storage as a heap.
 D. All of the above

57. The general algorithms exist for recognition and parsing is:
 A. $O(n^3)$ B. $O(n^4)$
 C. $O(n^2)$ D. $O(n^5)$
58. A critical section in a program segment:
 A. Which should run in a certain specified amount of time
 B. Which avoids deadlocks
 C. Where shared resources are accessed
 D. Which must be enclosed by a pair of semaphore operations, P and V
59. Which of the following derivations does a top-down parser use while parsing an input string? The input is assumed to be scanned in left to right order?
 A. Leftmost derivation
 B. Leftmost derivation traced out in reverse
 C. Rightmost derivation
 D. Rightmost derivation traced out in reverse
60. Binary number that are multiples of four is:
 A. $(0/1) * 00$ B. 100
 C. $1(0/1) * 00$ D. $11(1) * 00$
61. When the projection angle is chosen so that $\tan \leq 2$, the resulting view is called:
 A. Cavalier projection
 B. Orthographic projection
 C. Cabinet projection
 D. All of the above.
62. Gouraud shading is also known as:
 A. Flat shading method
 B. Intensity interpolation method
 C. Normal vector interpolation method
 D. Constant intensity shading method
63. In raster scanning, what is false?
 A. Vertical retrace is at the end of line scanning and horizontal retrace at the end of the frame.
 B. The refreshing rate of 60 frames per second is also called as 60 Hz.
 C. Each frame display whether in one pass/two pass is also termed as non-interlaced/ interlaced refresh.
 D. Frame buffer is also called as bitmap or pixmap.
64. The potential barrier at a PN junction is due to charge on either side of the junction. These charges are:
 A. Minority carrier
 B. Majority carrier
 C. Both minority and majority carrier
 D. Fixed donor and acceptor ions
65. What is reverse recovery time in diode?
 A. Time required for a diode to switch from off to on.
 B. Time required for a diode to switch from on to off.
 C. Time required for a diode to undergo reverse saturation effect.
 D. Time required for a diode to change from forward bias to reverse bias.
66. The power factor is:
 A. Ratio of apparent power to average power
 B. Simply $\sin \beta$, where β is the angle by which the voltage leads the current
 C. Ratio of real power to the apparent power
 D. Simply $\cos \beta$, where β is the inverse power factor angle.

67. The volume-of copper required for an Ac transmission line is inversely proportional to:
- Current
 - Power factor
 - Voltage
 - Both B and C
68. The common emitter forward amplification factor (β_{dc}) is given by:
- I_C/I_E
 - I_E/I_B
 - I_C/I_B
 - I_B/I_E
69. An amplifier power is changed from 10 watts to 20 watts. The equivalent dB gain will be:
- 2dB
 - 6dB
 - 3dB
 - 10dB
70. A major advantage of an emitter follower is that it provides.
- Maximum gain
 - Maximum output impedance
 - Maximum efficiency
 - Maximum distortion
71. Synchronous data transmission uses.
- Start pulses
 - Common clock pulse at transmitting and receiving
 - Stop pulses
 - None of the above
72. Poynting vector of an EMW gives the:
- Intensity of electric field
 - Intensity of magnetic field
 - Rate of energy flow
 - Direction of polarization
73. The propagation factor for dry air is 98% of C what is this, in miles/second?
- 172,282 miles/s
 - 182,282 miles/s
 - 122,322 miles/s
 - 180,444 miles/s
74. Probability of error or bit error rate (BER) is given by:
- $P(e) = Q(A/\sigma)$
 - $P(e) = Q(\sqrt{SNR})$
 - Both (A) & (B)
 - None of the above
75. High frequency long distance communication is achieved due to:
- Ground reflection
 - Inverted reflection
 - Ionospheric reflection
 - Tropospheric reflection
76. Expert systems (ES) has been applied to problem domains as:
- Operation Research/management Science
 - Statistical analysis
 - Selecting a multiple comparison procedure (MCP)
 - All of the above
77. Inference engine is used in:
- Expert system
 - Artificial intelligence
 - Neural network
 - Cryptography
78. Which statement is false?
- Not all internet servers are part of the World Wide Web
 - The default port number for HTTP is 80
 - HTTP responses always return HTML
 - None of the above

79. Digital signature is:

- A. Scanned image of normal signature
- B. Digitally protected form of analog signature
- C. Additional encrypted verifiable code attached with normal data content
- D. None of the above

80. Internet-like network between a company and its business partners. Select the best fit for the answer:

- A. Bandwidth alternatives
- B. Inter organizational networks
- C. Witching alternating
- D. Extranets

ANSWER SHEET

1.D	2.B	3.A	4.C	5.C	6.B	7.C	8.C	9.C	10.C
11.D	12.C	13.B	14.C	15.B	16.D	17.D	18.C	19.C	20.D
21.A	22.C	23.D	24.A	25.C	26.B	27.B	28.D	29.D	30.A
31.D	32.D	33.C	34.B	35.D	36.D	37.A	38.A	39.C	40.B
41.D	42.A	43.A	44.A	45.D	46.C	47.C	48.A	49.C	50.B
51.C	52.B	53.B	54.A	55.A	56.A	57.A	58.C	59.A	60.B
61.A	62.B	63.A	64.D	65.D	66.A	67.D	68.C	69.C	70.D
71.B	72.C	73.B	74.C	75.C	76.D	77.A	78.C	79.C	80.D